

LearnSMTH x SpaceX Inspired Experiential STEM Learning

"Turning Curiosity Into Real-World Innovation"

SRM Vadapalani Incubation Center Pitch

Presenter: BBA Final Year Student



News Clip: SpaceX Starship Test

A recent SpaceX Starship test flight made headlines — a high-altitude launch with simulated satellite deployment and a controlled splashdown. LearnSMTH used this real-world event as a springboard for student learning.

Class Activity: Guided news-reading and timeline reconstruction.

Hands-on Lab: Build simple rocket models to explore Newton's laws and aerodynamics.

Design Challenge: Teams design a sensor payload for community monitoring.

Data Skills: Interpret launch telemetry, times, and basic charts.

SpaceX's Starship flight most targets in pre-IPO

Starbase, Texas: SpaceX completed a largely successful test flight of its next-generation Starship rocket on Friday, deploying a clutch of mock satellites and executing a controlled splashdown in the Indian Ocean in a high-stakes debut of the newly upgraded vehicle as Elon Musk's company prepares to go public.

Starship, which SpaceX has spent \$15 billion developing as a fully reusable spacecraft, is critical to Musk's goals of cutting launch costs, expanding his Starlink business and pursuing ambitions ranging from deep-space exploration to orbital data centers — all factored into his targeted \$1.75 trillion IPO valuation.

Friday's launch marked SpaceX's 12th test flight of a Starship prototype since 2023 and the first of its V3 iteration, a major upgrade of both the cruise vessel and its Super Heavy booster, as well as the first



This screenshot made from a SpaceX live feed shows Starship 39 rocket landing in the Indian Ocean on Friday

blast-off from a launch pad specially designed for the new rocket.

The towering vehicle, consisting of the upper-stage Starship astronaut vessel stacked atop a Super Heavy booster rocket, blasted off around 5:30pm (2230 GMT) from SpaceX facilities in Starbase, Texas, on the Gulf of Mexico near Brownsville. A live SpaceX webcast showed the rocketship,

over 40 stories tall, launch tower as cluster of 33 Rapid

red to life in a ball. Minutes later, the vehicle to soar on, despite the loss of one, then release its payload before atmospheric re-entry. Its flight hour in all. The Heavy came down about six nautical miles off, as expected, but the rocket failed to complete its boost-back burn separating from the Starship. SpaceX said it would not attempt a recovery or the Starship if all else went as

How Kids Learn

We turn the clip into an integrated learning module: storytelling, role-play (engineers & reporters), measurement labs, science experiments, chemistry demos, simple coding for sensors, teamwork, and reflection to connect media with STEM concepts.

The Problem With Traditional Learning

× Current Limitations

- Memorization-heavy learning
- Low curiosity retention & lack of application
- Fear of failure in STEM subjects
- STEM disconnected from real life

Our Core Belief

Children learn deeply when they are **emotionally engaged**, solving meaningful problems, collaborating, and building something real.

*"What if children didn't just study science... but **build** solutions inspired by SpaceX?"*

Meet The Spark Squad

Children emotionally connect with recurring STEM heroes. This creates continuity, engagement, learning ideas, and long-term retention.





Mission 04: The Community Satellite Challenge

Story Setup

A cyclone warning system failed near OMR Chennai.

The Spark Squad is challenged to design a mini community rocket + satellite mission that can help monitor:

Flooding and Rainfall

Heat

Emergency alerts

Big Question: "How can young innovators use technology to help communities?"



Turning Curiosity Into Core STEM Knowledge

Physics

Newton's Third Law: Rocket pushes gases down → rocket moves up.

Aerodynamics: Why rocket shape matters.

Students EXPERIENCE physics instead of memorizing formulas.

Chemistry

Combustion: Fuel + Oxygen reactions.

Energy: Heat generation & pressure build-up.

Focus on safe, visual and age-appropriate understanding.

Math

Measurement, Angles, Time
Distance calculation.

Emotional Shift: Children stop asking "Why do we learn math?"

Instead: "We need math to improve our rocket."

Why SRM Vadapalani? Shared Vision

Opportunity For Collaboration

- Incubation mentorship
- Prototype support & PCB development guidance
- Maker lab collaboration
- Engineering student volunteers

Building India's next generation of innovators.

The End Goal

*"What If Chennai Apartments Became
Innovation Labs?"*

*"Today's mission-based learners can become
tomorrow's innovators."*